

DESIGN AND IMPLEMENTATION OF A CENTRALIZED LOG MONITORING AND THREAT DETECTION SYSTEM USING ELK STACK

Submitted By
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INTRODUCTION

This project implements centralized Windows Security Event monitoring using Elasticsearch, Logstash, Kibana and Winlogbeat. Windows Event Logs are collected, processed, indexed and visualized through Kibana dashboards.

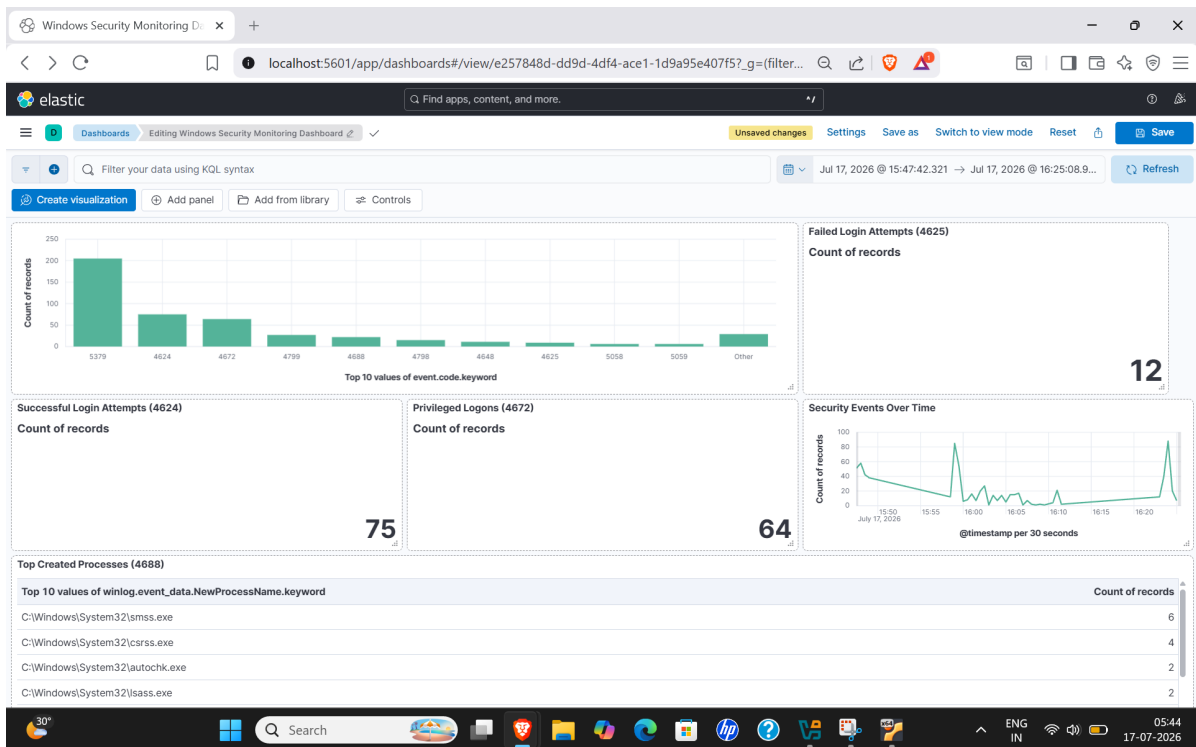
OBJECTIVES

- Centralize Windows logs
- Process logs with Logstash
- Store logs in Elasticsearch
- Visualize events in Kibana
- Demonstrate a SOC workflow

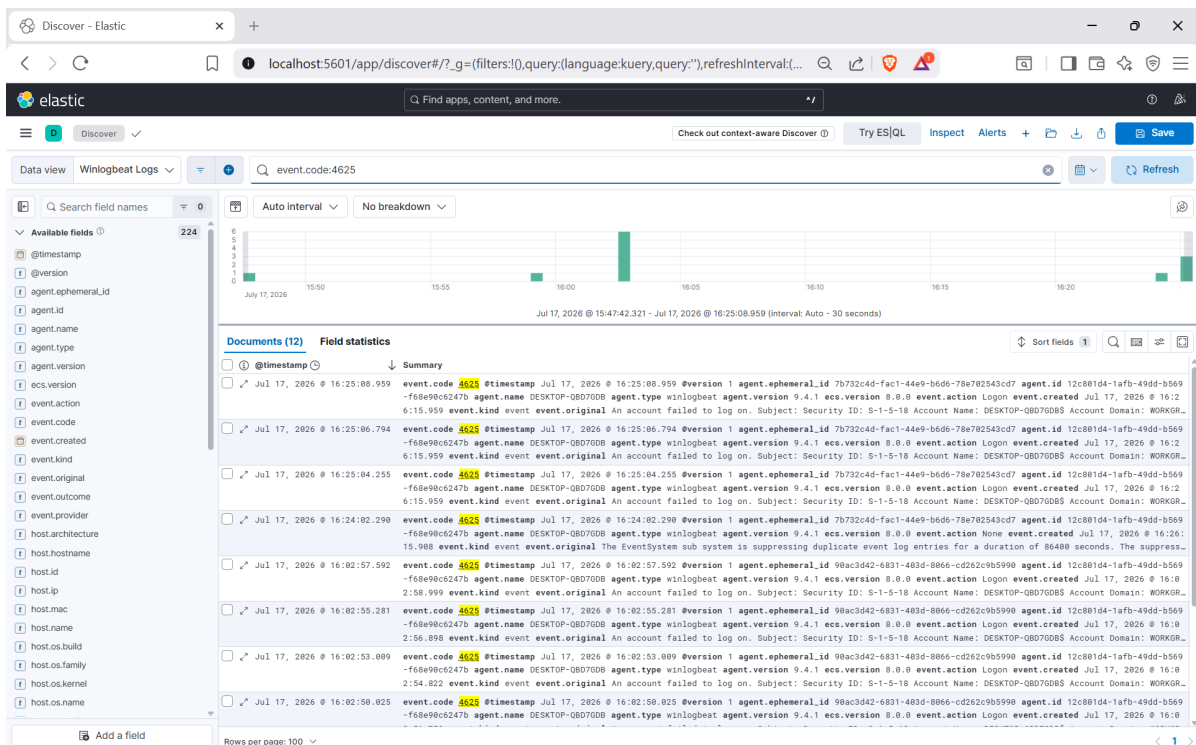
SECURITY EVENTS MONITORED

4624 Successful Login
4625 Failed Login
4672 Privileged Logon
4688 Process Creation

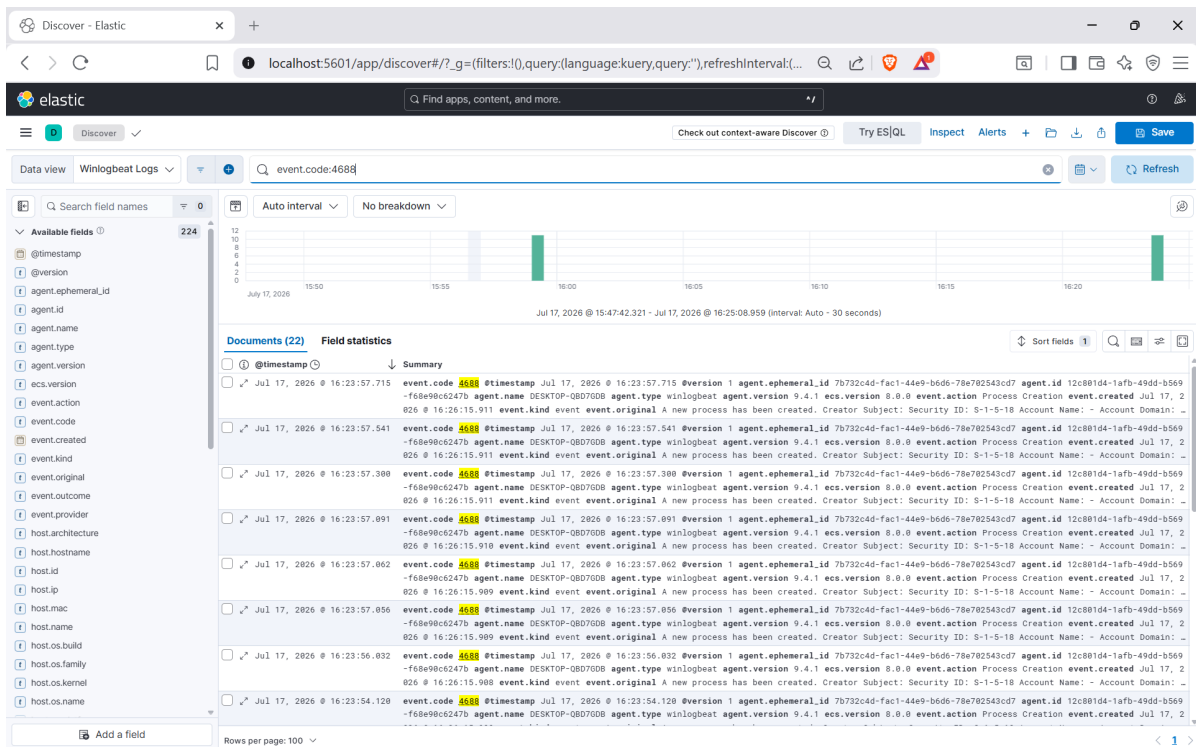
Kibana Security Dashboard



Failed Login Events (4625)



Process Creation Events (4688)



CONCLUSION

The project successfully demonstrates a working ELK-based SIEM pipeline with Kibana dashboards for monitoring Windows Security Events.

FUTURE SCOPE

Future work includes alerting, ML anomaly detection, Wazuh integration and additional log sources.